Technology Stack

Date	23 October 2026
Team ID	[Your Team ID]
Project Name	TravelMate
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (React, Angular, HTML/CSS)	React + Vite, JavaScript (JSX), CSS, Leaflet, jsPDF	Component-based reusable UI, fast Vite builds, responsive layouts, interactive maps and client-side PDF export.
2	Backend / Server-Side (Node.js, Python/Django, Java)	Python, FastAPI, Uvicorn	Fast, lightweight API framework with data validation through schemas; suits AI service integration.
3	Database / Data Storage (MySQL, MongoDB, PostgreSQL)	Supabase (hosted PostgreSQL and Auth)	Managed database with built-in authentication, so user sign-in and saved trips need no separate server.
4	Cloud / Hosting / Deployment (AWS, Azure, Heroku, Docker)	Vercel (frontend), Render (backend)	Simple deployment from GitHub; Vercel suits static React builds and Render runs the Uvicorn server.
5	Version Control & CI/CD (GitHub, GitLab, Jenkins)	Git and GitHub	Version history, safe collaboration and a source for deployments.
6	Third-Party APIs / Other Tools (Stripe, Twilio, Google Maps)	Groq API, OpenStreetMap, Nominatim, Google Maps links	Groq generates the AI itinerary; OpenStreetMap and Nominatim draw map tiles and convert place names to coordinates; Google Maps links give directions.